KAPASA MAKASA UNIVERSITY

CYS 339: Fraud and Detection Technologies Quiz

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Total Marks: 40 Duration: 50 minutes

Section A: Multiple Choice (10 Marks): Answer ALL questions. Each question carries 1 mark.

1. Which of the following is NOT a type of quantitative data analysis?

a) Descriptive Analysis b) Diagnostic Analysis

c) Contextual Analysis d) Predictive Analysis

2. Financial ratio analysis in cybersecurity is primarily used to:

a) Detect malware signatures b) Measure ROI on security investments

c) Encrypt sensitive data d) Generate phishing emails

3. A favourable variance in cybersecurity indicates:

a) Actual costs > Expected costs b) Actual costs < Expected costs

c) No change in costs d) Increased breach likelihood

4. Which function combines data from different tables based on a common key?

a) Filter b) Append c) Join d) Pivot

5. Aging analysis is most effective for detecting:

a) Zero-day exploits b) Dormant account reactivation

c) Encryption failures d) Firewall misconfigurations

6. What is the primary limitation of using averages in fraud detection?

a) They are computationally expensive b) They can be skewed by outliers

c) They only work with qualitative data d) They require manual updates

7. Which technique divides data into homogeneous subgroups for analysis?

a) Regression b) Stratification c) VLOOKUP d) Trend analysis

8. A key disadvantage of filtering in cybersecurity is:

a) Increased data volume b) Risk of over-filtering and missing subtle attacks

c) Inability to handle numerical data d) High cost of implementation

9. Which of the following is a qualitative data analysis method?

a) Sorting b) Behavioral analysis c) Ratio calculation d) Variance analysis

10. What does a high false-positive ratio in IDS alerts suggest?

a) Effective threat detection b) Poorly tuned detection thresholds

c) Encrypted traffic d) Low attack volume

Section B: True or False (10 Marks)

Indicate whether the statement is True (T) or False (F). Each question carries 1 mark.

1. Quantitative data analysis focuses on "why" and "how" rather than "what" and "how much."

2. Aging analysis can detect insider threats by flagging dormant accounts that suddenly become active.

3. A high false-positive rate in IDS alerts always indicates superior threat detection.

4. VLOOKUP can search for values to the left of the lookup column.

5. Behavioral analysis is a qualitative method that detects anomalies based on deviations from normal patterns.

6. Financial ratios in cybersecurity are only useful for measuring malware detection rates.

7. Stratification reduces noise in datasets by grouping similar events, but may hide outliers.

8. Predictive analysis uses machine learning to forecast future threats based on historical data.

9. Append operations combine tables horizontally using a common key.

10. Qualitative analysis is more effective than quantitative analysis for detecting zero-day exploits.

Section C: Matching (10 Marks)

Match each term (1-10) with its correct definition (A-J). Each match carries 1 mark.

Term	Definition
1. Join	A. Examines geographic patterns in threats
2. VLOOKUP	B. Organizes data into meaningful order
3. Stratification	C. Combines tables based on a common key
4. Aging Analysis	D. Searches vertically for a value in a table
5. Spatial Analysis	E. Divides data into homogeneous subgroups
6. Trend Analysis	F. Tracks time elapsed since an event
7. Sort	G. Identifies patterns over time
8. Pivot Table	H. Summarizes and cross-tabulates data
9. Regression	I. Models relationships between variables
10. Behavioral Analysis	J. Detects anomalies in user/system behaviour

1(……). 2(….) 3(…..) 4(…..) 5(…..) 6 (….) 7(…..) 8(…..) 9(….) 10(….)

Section D: One-Word Answer/Phrase (10 Marks)

Answer in ONE word or a short phrase. Each question carries 1 mark.

1. What type of analysis is used to predict future cyber threats based on historical data?

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2. Which function stacks datasets vertically?

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3. What is the primary disadvantage of using averages in fraud detection?
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4. Name a qualitative data analysis method that examines attacker motives.

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5. What does a favourable variance indicate in cybersecurity costs?

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6. Which statistical measure is most skewed by outliers?

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7. What is the main purpose of stratification in cybersecurity?

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8. Which analysis type is best for detecting seasonal attack patterns?

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9. What is the key limitation of filtering in threat detection?

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10. Which technique is used to group vulnerabilities by severity?

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